Interpretation of Frequency-Domain Airborne Electromagnetic Data Based on the Deep Neural Network Incorporating Topographic Information

Minkyu Bang, Soon Jee Seol, and Joongmoo Byun

***Abstract*— Deep neural networks (DNNs) have recently been used to interpret frequency-domain electromagnetic (EM) data; therefore, vast amounts of information can be rapidly interpreted. However, in airborne surveys that include mountainous regions, severe topographic changes distort the EM data, and they can lead to unreasonable interpretations. DNN-based inversions that reduce the effects of topographic changes have not yet been proposed, whereas some conventional inversion techniques can correct topographic distortions. Since DNN-based interpretation needs various training datasets and its performance depends on the characteristics of training dataset, it is important to generate various airborne EM (AEM) data and resistivity model pairs. However, it is almost impossible to derive the EM responses of 2-D or 3-D resistivity models incorporating diverse topographic patterns, because such models require many variables and computation costs are limited. Therefore, we suggested the pseudo-1-D interpretation of EM data, which can consider the topographic changes using EM responses and topographic information at three neighboring data points. We include topographic information in DNN training as slope angle between the data points. Compared with conventional inversion, our trained DNN model recovers the synthetic resistivity models more reasonably. In addition, we used the trained DNN model to evaluate a real AEM dataset from Southcentral Alaska. The trained DNN provided reasonable interpretation results despite that the data were acquired at mountainous area with rough topography. Since the trained DNN model can provide predictions rapidly, the interpretation time was drastically reduced. We believe that this research can increase the practical field applicability of AEM survey.**

***Index Terms*— Deep neural network (DNN), frequency-domain airborne electromagnetic (AEM), pseudo-1-D method, topography.**

I. Introduction

GIVEN the advances in airborne electromagnetic (AEM) data acquisition systems, EM surveys covering large areas have become possible. Unlike land EL surveys, AEM surveys usually collect a vast amount of dataset; therefore, most of the AEM data interpretations were performed using 1-D approaches, such as conductivity depth transformation [1], S-inversion [2], Zodhy's method [3], and layered-earth inversion [4] instead of 2-D or 3-D inversions, which takes a lot of time and computational costs. However, even though 1-D approach is the simplest and fastest interpretation methods, conventional inversions have a limitation to accommodate millions of data. Furthermore, the results are highly dependent on the inversion parameters (the resistivity of the initial model, inversion block size, and weighting factors), all of which are set by (well trained and experienced) human professionals.

Recently, with the advances in artificial intelligence, deep neural network (DNN)-based data processing and interpretation have been used by geophysicists to minimize complex computations and human interventions. Likewise, DNN-based AEM data interpretation methods have been developed. Noh et al. [5] compared the interpretations of 1-D Gauss–Newton inversion and 1-D DNN-based inversion; the latter dramatically reduced the interpretation time and showed similar result of conventional inversion. Li et al. [6] developed a 1-D convolutional neural network (CNN) that recovered 1-D resistivity structures. Asif et al. [7] presented an NN that replaced the forward-modeling and Jacobian calculations of conventional inversion, reducing the need for human intervention and the calculation time. Bang et al. [8] developed a DNN-based 2-D inversion method using a recurrent neural network (RNN) and 2-D inputs and labeled pairs. Through 2-D RNN inversion, they successfully showed clear images of conductive anomalies with high resolution.

However, suggested DNN-based methods are early stage trials for practical applications, because they cannot consider topographic variations during the inversion process. Since many airborne surveys have been performed at mountainous area, the EM responses are likely to be distorted by topographic changes. The importance of topographic information during interpretation has been emphasized by several studies. Oh et al. [9] showed that EM responses were distorted by topographic changes, and that 2-D inversion of distorted data (without topographic information) provided unreasonable results from a land EM survey. Compared with land EM responses, even if AEM responses were less affected by topographic changes, inversion of AEM data without considering the topography also provided unreasonable results. Ghari et al. [10] proved that the inversion of AEM data incorporating with topography information could provide reasonable results and showed satisfactory performance to obtain

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The authors are with the Earth Resources and Environmental Engineering (EREE) Department, Hanyang University, Seoul 04763, South Korea (e-mail: wbmaicon13@gmail.com; ssjdoolly@hanyang.ac.kr; jbyun@hanyang.ac.kr).

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a clear image of conductive anomalies. Therefore, to improve the practicality of DNN-based interpretations, topographic information is required during training process. However, this is very difficult, because the topographic patterns can be represented in 2-D or 3-D. Thus, data and labeled pairs must be prepared using 2-D or 3-D resistivity models and their EM responses. Even in 2-D situations without topographic changes, the resistivity models require a lot of parameters [8]. Thus, it is almost impossible to prepare a training dataset incorporating the topographic patterns of 2-D or 3-D resistivity models given the current computational environment.

In this article, we present an efficient form of 1-D DNN inversion that alleviates the topographic distortion of AEM data using only three neighboring data points and their topographic information. To overcome the complexity of topographic models in high dimensions, the training data were generated using a pseudo-1-D method [11] that produces three, consecutive, frequency-domain EM responses employing a bent, layered-earth model. This method represents the topographic changes as the slope angles from the left and right positions to the midpoint. Thus, many pseudo-1-D resistivity models (with various topographic patterns) can be generated (for training) via simple manipulation of the angles at both positions of the layered-earth models. The DNN inversion model learns the complex patterns of EM responses by changes in the topographic patterns and the resistivity models. The input data are constructed using the configuration of DIGHEM-V system, and the developed 1-D DNN inversion algorithm will be tested using both synthetic data and field data acquired over a mountainous area of Farewell, Alaska. To evaluate the effects of topography information during training process, the results not only of 1-D DNN inversion (without topographic consideration) but also 1-D Gauss–Newton inversion are compared with those of our 1-D DNN inversion method that incorporates topographic changes. Finally, we present the entire prediction results of a field AEM survey. Our method is computationally efficient and corrects topographic distortion during AEM data interpretation.

II. Field Data Analysis

In this study, we aim to develop an efficient DNN inversion algorithm that interprets frequency-domain AEM data acquired using the DIGHEM-V [12] that has been widely used for mineral exploration and subsurface characterization of field sites. We selected a dataset acquired in the Yentna and McGrath mining districts of Southcentral Alaska (Fig. 1). Airborne surveys were performed from September 12 to November 3 in 2012 and from June 29 to September 27 in 2013 to detect mineral deposit. Through the airborne survey, data were acquired at three million points over a length of 7187.5 km and an area of 2544.9 km^2.

As is well known, DNN performance is highly dependent on the construction of training dataset [14]. Therefore, we systematically analyzed the specifications of DIGHEM-V system (frequencies, transmitter–receiver separations, sampling rates, and sensor height ranges) before generating the training dataset.

Fig. 2 shows a schematic of the DIGHEM-V system operating in the horizontal coplanar (HCP) and vertical coaxial (VCA) configurations. Usually, only the quadrature component is used during EM interpretation, because the in-phase component contains negative outliers attributable to magnetic anomalies [15] and the quadrature component correlates strongly with conductive anomalies. In addition, VCA datasets are much noisier than HCP datasets. Thus, we focused on the quadrature component of the HCP dataset. The HCP configuration featured a vertical magnetic dipole (VMD) as a source; two receiver coils measured the ratios of

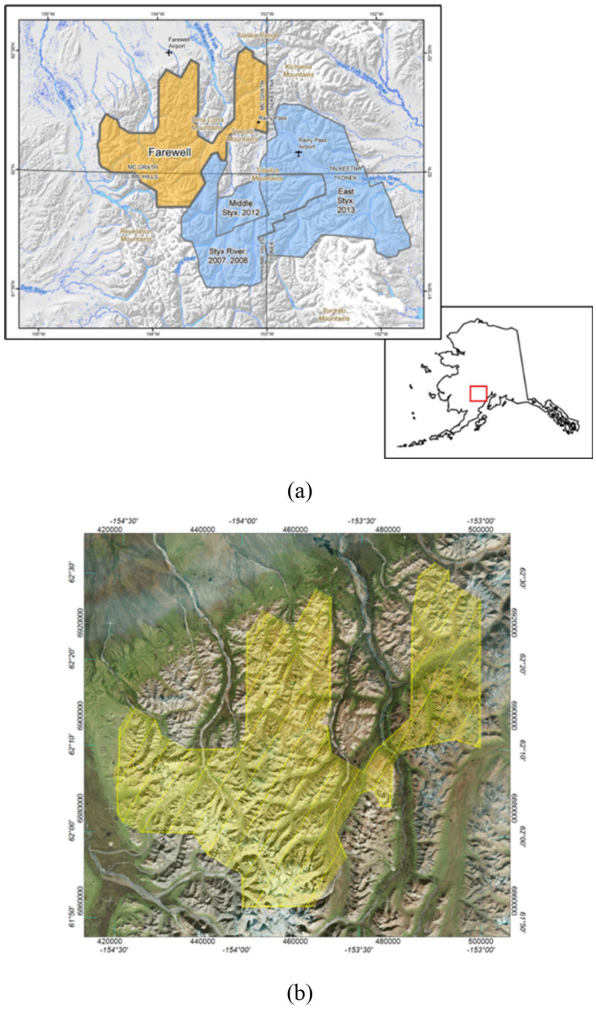


Fig. 1. (a) Location of the Farewell electromagnetic and magnetic airborne geophysical survey in Southcentral Alaska. (b) Yellow lines are the airborne survey lines (both images from Burns et al. [13]).

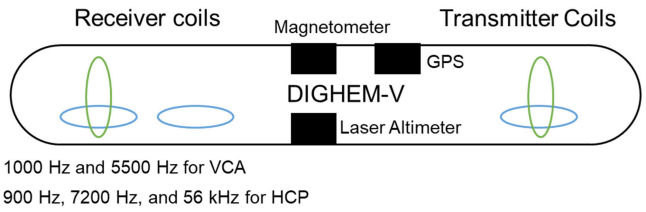


Fig. 2. Schematic of the DIGHEM-V system that measures EM responses in the HCP and VCA configurations over multiple frequencies. The system records spatial co-ordinates and altitudes using a GPS and a laser altimeter, respectively. The system also measures the residual magnetic field (using a magnetometer).

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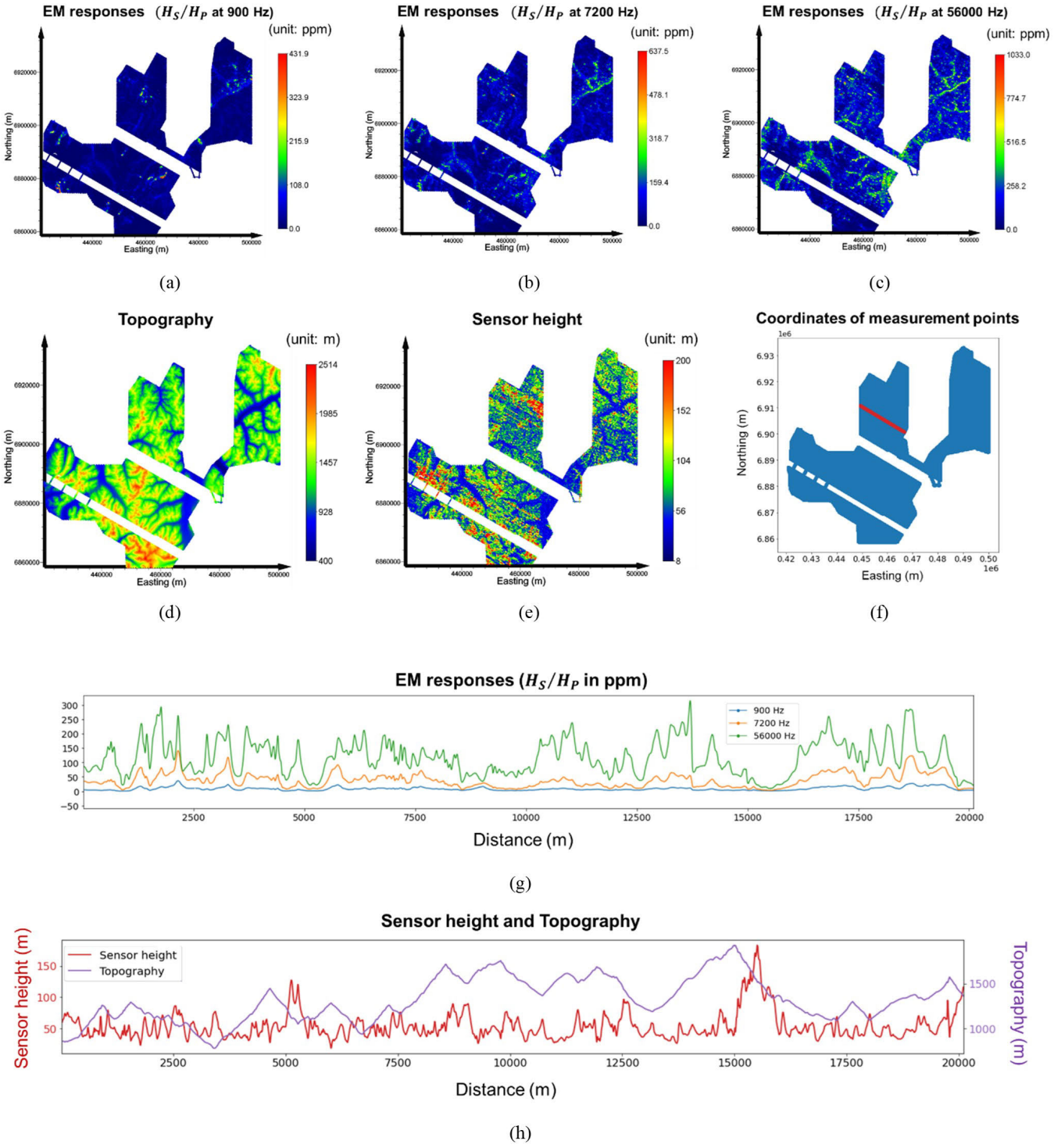


Fig. 3. Map views of the entire dataset acquired over Southcentral Alaska using the DIGHEM-V system. (a)–(c) Quadrature component maps of the EM responses derived using the HCP configuration for each frequency. (d) Topographic map. (e) Recorded sensor height map. (f) Co-ordinates of the measurement points. To analyze the data characteristics in detail, (g) EM responses and (h) topographies and sensor heights are plotted [with the survey line marked in red in (f)].

the vertical secondary and primary magnetic fields (H_S/H_P) (in ppm) at three frequencies (900 Hz, 7200 Hz, and 56 kHz) (Fig. 2).

During the survey, the system records not only the EM responses but also co-ordinates, sensor heights, and digital terrain model (DTM) information. Fig. 3 shows the dataset collected by the airborne survey, and as marked red line in Fig. 3(f), each line consists of vast amount of data series [Fig. 3(g) and (h)]. As we can see in the topographic variations in this survey area [Fig. 3(d) and (h)], the survey was performed over the mountainous area with severe topographic change; therefore, the height of AEM system was also highly varied, and EM responses showed very fluctuated responses [Fig. 3(g)].

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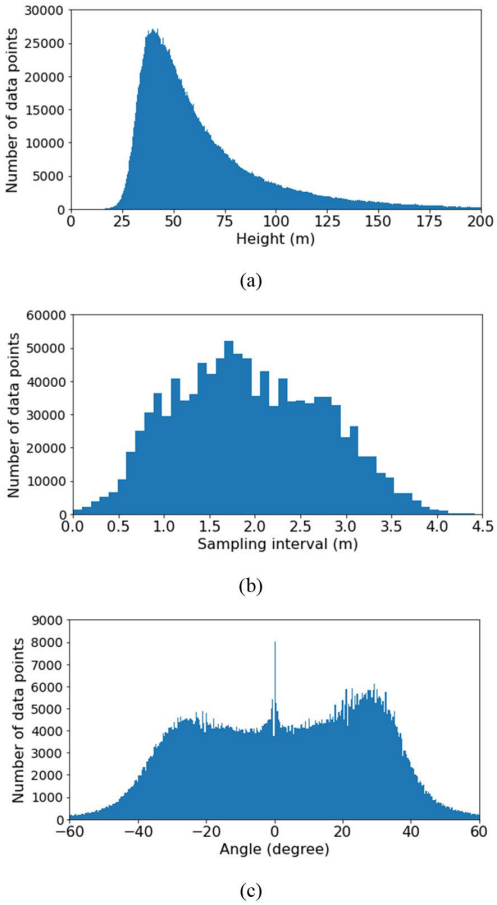


Fig. 4. Histograms of the field sensor heights, lateral distances, and topographic changes. (a) and (b) Sensor heights and lateral distances between data points (respectively). (c) Slope angles calculated by the arctangent function using the vertical and lateral distances between data points.

For more detailed inspection of survey parameters, we drew histograms of the sensor heights, lateral distances between points, and slope angles between points for all points (Fig. 4). The slope angle indicates the topographic change between pairs of data points and is calculated by applying the arctangent function to the lateral distance and the elevation difference between data points. Given the severe topographic changes, the slope angle generally ranges from $-30°$ to $+30°$ and even attains $\pm 60°$ [Fig. 4(c)]. The most common sensor height is around 40 m, and the average sampling interval is 2 m. To generate training EM responses that reflect the field data accurately, these parameters must be considered during forward modeling. We now discuss how to use these survey parameters into the forward-modeling stage in next paragraph.

III. Generation of a Training Dataset via Pseudo-1-D Modeling

Theoretically, it is impossible to correct topographic effects in 1-D interpretation, because such topographic changes can be represented in 2-D or 3-D geometries. However, the construction of 2-D resistivity models with topographic changes is almost impossible, because many variables are required to describe 2-D resistivity models, such as anomalies, and their resistivities and spatial shapes; computation of the EM responses is extremely costly [8]. Therefore, it is necessary

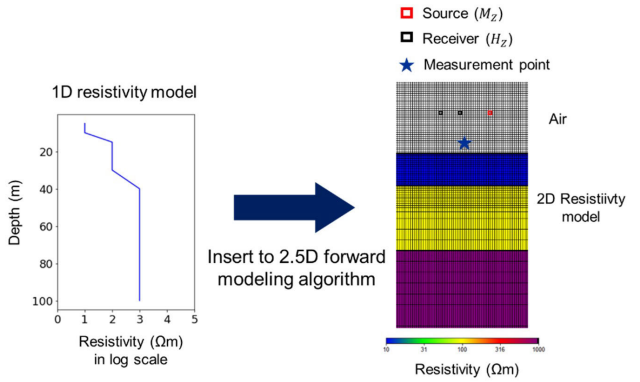


Fig. 5. Forward-modeling simulating the EM responses acquired by DIGHEM-V. Left: example (synthetic) of the 1-D resistivity model. Right: 2-D representation of the model used to generate EM responses via the 2.5-D modeling algorithm. The configuration of the source (red dot) and receivers (black dots) and the measurement point (star) are superimposed on the resistivity model.

to design a simple systematic way, which can adapt the topographic changes to generate plenty amount of training dataset for enough training of the DNN inversion model.

We proposed the "pseudo-1-D" modeling, which produce three consecutive EM responses for bent layered structures. We first constructed 1-D resistivity models [Fig. 5(a)] and inserted them into a 2.5-D forward-modeling algorithm [9] [Fig. 5(b)]. To simulate the HCP configuration of DIGHEM-V, two receivers (with different offsets) were placed on a measurement point for a VMD source.

Next, we set three consecutive measurement points and varied the slope angles of the left and right positions centering to a measurement point [Fig. 6(a)]. Through the field data analysis in Section II, the lateral sampling intervals were mostly 1–3 m; we, thus, set the lateral distance between measurement points to 2 m (the average). In addition, as most slope angles ranged between $\pm 30°$ and as steep slope angles (over $\pm 40°$) can cause the numerical instability during the forward modeling, we excluded the steep angles exceed $\pm 40°$. Finally, as shown in Fig. 6(b), we generated 81 topography patterns by combinations of nine angle values of left and right positions and applied 81 topography patterns to each resistivity model.

The forward-modeling algorithm simulates topographic patterns by moving the nodal points of resistivity models, and its accuracy was verified by Oh et al. [9] comparing with the analytic solutions of EM1D [16]. The EM responses incorporating the topographic changes of the 1-D resistivity model were calculated by arranging the node co-ordinates of the left and right positions along the slope angles.

As shown in Fig. 3(d), as the field topography shows significant ranges from 400 to 2500 m, it was very difficult to maintain a constant sensor height during the airborne survey. Hence, Fig. 4(a) shows that the data were collected over a wide range of altitude (20–200 m). To simulate this, the EM responses must be calculated at various heights. However, we did not consider sensor heights above 70 m during forward modeling, because EM responses at all three frequencies are then very low regardless of any change in subsurface resistivity due to sensitivity declines rapidly with height [17], [18]. For the training dataset, we calculated EM responses at sensor

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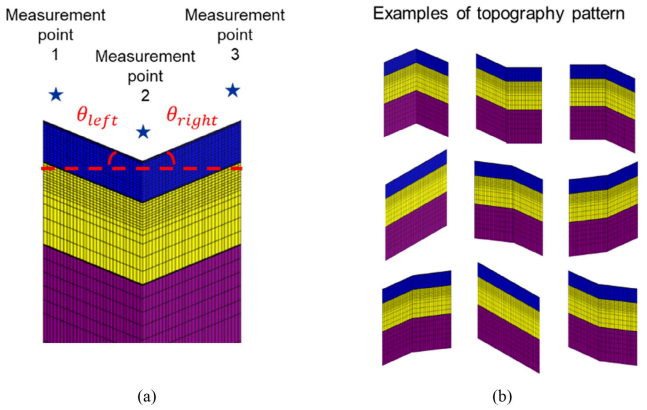


Fig. 6. Schematic of the pseudo-1-D method that calculates the EM responses of three measurement points including topographic changes. (a) Topographic patterns were generated by changing the slope angles of the left and right positions that were centered on the middle point of three consecutive data points. (b) Small resistivity models show some of the generated topographic patterns.

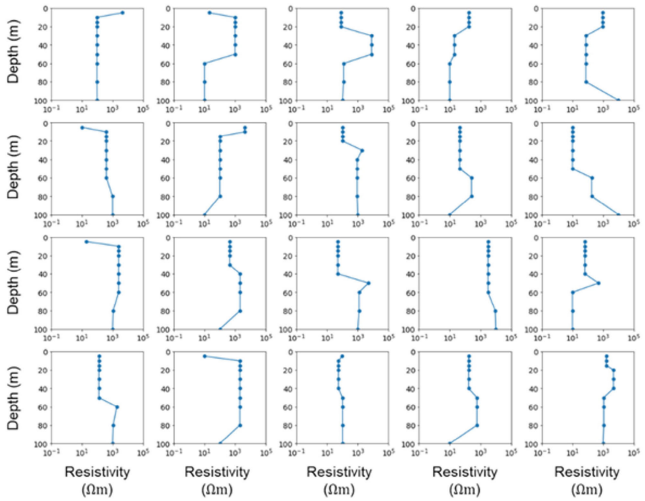


Fig. 7. Examples of 1-D resistivity models (basic models used to simulate 81 topographic patterns).

heights of 20, 25, 30, 35, 40, 45, 50, 55, 60, and 70 m. As EM responses are insensitive to changes above 60 m [8], we omitted the 65-m data to reduce modeling time.

DNN model performance is critically dependent on the characteristics of training dataset; therefore, it is essential to generate various resistivity models and calculate their EM responses. We generated 6300 1-D resistivity models reflecting various situations, and Fig. 7 shows some examples. The resistivity models were generated using the following rules: 1) the maximum depth was 100 m, because the normalized sensitivity [19] of the half-space model has been intensively studied from the surface to a 100-m depth [8]; 2) 1-D resistivity model consists of single-layered model to four-layered model; 3) the boundaries can lay at 5-, 10-, 15-, 20-, 30-, 40-, 50-, 60-, and 80-m depth; and 4) the resistivity ranged from 1 to 10 000 Ωm to simulate various subsurface conditions. In order to facilitate the reproducibility of 1-D resistivity models, algorithm for model generation is described in the Appendix.

We set maximum horizontal range to 200 m, because a single data point has a sensitivity for 200 m in horizontal direction [8]. In addition, to reduce the numerical errors and obtain reliable data, we set horizontal mesh size to 1 m and set vertical mesh size from 0.5 m (at surface mesh) to 5 m (at deepest mesh), because the sensitivity of conductance (conductivity × thickness) at deeper depth decreases. Totally, we generated 200 × 41 elements for subsurface model and added increasing logarithmic boundary elements for air and the three ends of the model. In addition, Neuman boundary condition was applied to the ends of boundary grids. As a result, totally 5 103 000 (81 topographic patterns, ten sensor heights, and 6300 resistivity models) input and label pairs were prepared to train our DNN inversion model, which, thus, learns patterns in the sensor height and EM responses, and the topographic changes, of three neighboring data points and predicts the 1-D resistivity of the midpoint.

IV. Design of a 1-D DNN Inversion Model

To train a DNN model incorporating topographic changes, we constituted the features of input data with height, topographic information (the slope angles), and EM responses of three consecutive measurement point sets. The topography information for each middle point was zero, because the topographic changes are the slope angles of the left and right points to the middle point.

Among the various DNN architectures, CNNs have been widely used to reconstruct resistivity structures in EM data inversion [6], [20]. CNNs handle image data effectively; a convolution operator considers the relationships between neighboring pixels. However, as we input three different data types (heights, topographies, and EM responses) in sets of three, we used an RNN, which has an advantage to handle continuous dataset. We employed long short-term memory (LSTM) [21] to solve the vanishing gradient problem of a conventional RNN. Compared with the conventional RNN, LSTM updates not only hidden state (h_t) but also cell state (c_t), which enables the network to learn long-term dependencies. If the data sequence (x_t) is given at the tth step, the hidden state (h_t) and the cell state (c_t) in a single LSTM unit consist of forget gate (f_t), input gate (i_t), and output gate (o_t) as follows:

$$h_t = o_t \odot \tanh(c_t) \tag{1}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \tag{2}$$
$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{3}$$
$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{4}$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{5}$$
$$g_t = \tanh\left(W_g x_t + U_g h_{t-1} + b_g\right) \tag{6}$$

where W, U, and b are parameter matrices and the parameter vector, respectively, and $\odot$ denotes the elementwise multiplication of vectors. In addition, we used bidirectional LSTM [22], which calculates hidden state in forward direction ($\vec{h}_t$) and backward direction ($\overleftarrow{h}_t$) as follows:

$$h_t = \left[\vec{h}_t, \overleftarrow{h}_t\right] \tag{7}$$

where [ ] operation denotes matrix concatenation. Since spatially adjacent EM responses can affect each other, bidirectional LSTM can learn both two-way characteristics.

We used not only a noise-free dataset, but we also added two different sets of noise-added datasets, both to prevent

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overfitting and to enhance DNN robustness to noise. We added up to 2 ppm of Gaussian noise (the reference level of the DIGHEM-V system). The total number of input–label pairs was 15 309 000, and we split the pairs into training (90%) and validation (10%) datasets, respectively.

In addition, to evaluate model performance, we combined the structure similarity index measure (SSIM) [23] and the mean absolute error (MAE; $L1$ loss)

$$\text{loss} = \alpha \text{MAE} + \beta(1 - \text{SSIM}) \tag{8}$$

$$\text{MAE} = \frac{1}{N}\sum_{i}^{N}|y_i - \hat{y}_i| \tag{9}$$

where y and $\hat{y}$ are the true and predicted labels, respectively, and α and β are the weighting factors for the two measures. The SSIM is

$$\text{SSIM} = \frac{(2\mu_y\mu_{\hat{y}} + c_1)(2\sigma_{y\hat{y}} + c_2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + c_1) + (\sigma_y^2 + \sigma_{\hat{y}}^2 + c_2)} \tag{10}$$

where μ_y, $\mu_{\hat{y}}$, σ_y, $\sigma_{\hat{y}}$, and $\sigma_{y\hat{y}}$ are the mean, standard deviation, and covariance of the true label y and the predicted label $\hat{y}$, respectively, and c_1 and c_2 are constants reflecting the dynamic ranges of the labels. Since EM response reflects conductance (conductivity × thickness) of subsurface structure and it uses the data at one position, 1-D inversion suffers more from nonuniqueness problem than 2-D or 3-D interpretation. In other words, the EM responses can be very similar when the resistivity models (at least) seem to be very different, but the conductances are similar.

Thus, we incorporated the SSIM (which measures the structural similarities of the predicted and true labels) into the loss function to recover not only the resistivity values but also structural trend of 1-D model simultaneously. In addition, SSIM values range from 0 to 1; a higher value indicates better prediction, and a lower MAE indicates better performance. Therefore, when combining both indices into a single loss function, we subtracted the SSIM from 1 as written in (8). The weightings of MAE and SSIM were set to 70% ($\alpha = 0.7$) and 30% ($\beta = 0.3$), respectively, to put slightly more weights on the minimizing misfit of resistivity values than matching the resistivity trends.

Our DNN model features nine bidirectional LSTM layers with 512 hidden units for each layer to accommodate such numerous numbers of input–label pairs. As well known, since geophysical inverse problem is finding the best solution solving nonlinear patterns between data–label pair, enough number of training parameters is needed to obtain satisfactory performance. Therefore, the DNN model requires a lot of training parameters to learn various data–label characteristics; hence, we built the DNN model using many layers. To verify that designed DNN architecture can learn various characteristics of training dataset and can prevent overfitting problem, we trained DNN with various number of hidden units and layers. We generated additional test dataset (1000 data–label pair) and estimated loss values using DNNs with different hyperparameters.

Fig. 8 shows the estimated loss of DNN along the number of layers (x-axis) and hidden units. As shown in Fig. 8, the minimum loss was 1.9% when DNN is built with nine LSTM layers and 512 hidden units. In the case of DNN

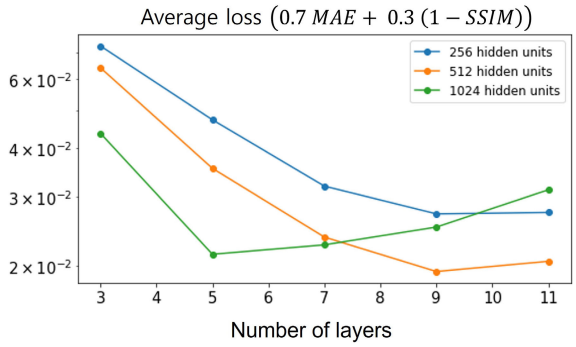


Fig. 8. Comparison of DNN performance to the test dataset. To find the best hyperparameters of DNN, DNN was trained by different numbers of layers (x-axis) with 256, 512, and 1024 hidden units (blue, orange, and green lines), respectively.

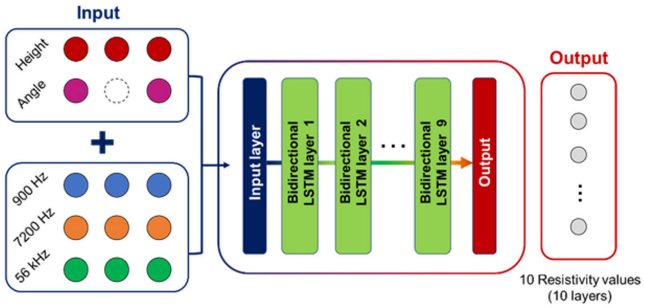


Fig. 9. DNN with bidirectional LSTM layers. The DNN predicts the ten values of 1-D resistivity using sensor heights, slope angles, and the EM responses of sets of three neighboring data points.

with 256 hidden units, all losses were higher than those of 512 hidden units. On the other hand, DNN with 1024 hidden units showed acceptable performance until five-layered model, but it was overfitted on the training dataset when more layers are added. Finally, we designed the DNN model to predict ten resistivity values of a 1-D resistivity structure (Fig. 9).

Other hyperparameters, such as activation functions, recurrent activation functions, and recurrent dropout, were tanh function, sigmoid function, and 0% recurrent dropout, respectively, because we used the CUDA 11.2 and cuDNN 8.1.1 environments (which accelerate computation), and they required those hyperparameter settings.

The Adam optimizer [24] was used to train the DNN model, and the learning rate decreased from $1e-03$ to $1e-06$ during training, which ceased at epoch 1200. The training and validation losses (the weighted sums of the MAE [Fig. 10(b)] and SSIM [Fig. 10(c)]) converged to 1.3 and 1.5%, respectively [Fig. 10(a)]. The DNN model was trained for 108 h, and the final algorithms were developed using Intel Fortran (oneAPI 2021), Python 3.8, and TensorFlow 2.7 employing an NVIDIA GeForce RTX 3090.

V. Numerical Experiments

To investigate the effects of topographic changes on measured EM data, we compared the EM responses (H_S/H_P) of three-layered models with [Fig. 11(a)] and without [Fig. 11(b)] topographic variations at 900 Hz, 7200 Hz, and 56 kHz [Fig. 11(c)]. The synthetic model features a conductive thin layer on a 1000-Ωm background and is of dimensions 600 and

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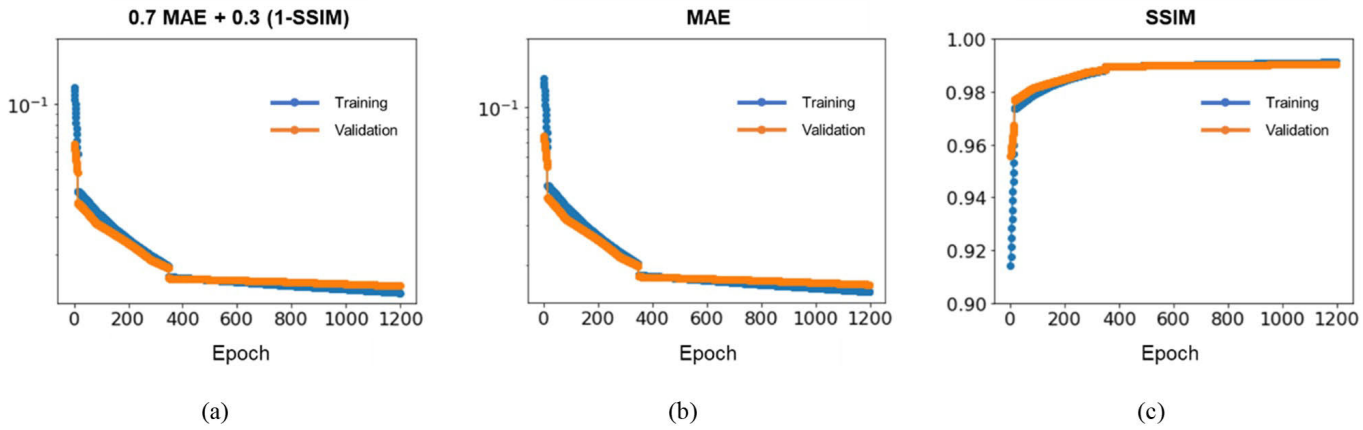


Fig. 10. History of DNN training. (a) Loss curves (the weighted sums of the MAE and modified SSIM for the training and validation data, respectively). (b) and (c) Training histories of the MAE and SSIM (training and validation data, respectively).

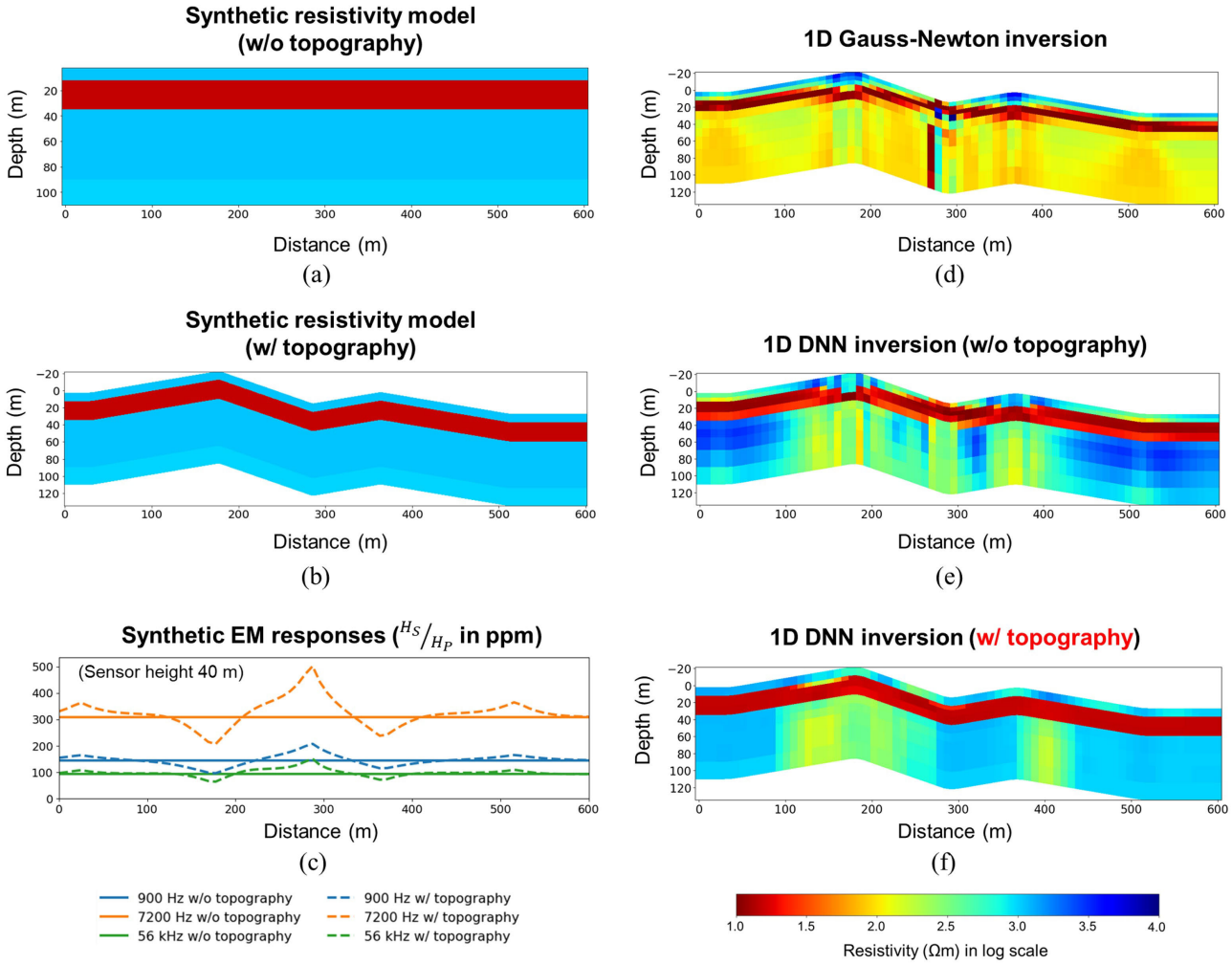


Fig. 11. Synthetic 2-D resistivity models (a) without and (b) with topographic changes, and (c) their synthetized EM responses. Dashed and bold lines in (c) represent the calculated EM responses of resistivity models (a) and (b), respectively, at three frequencies. The differences in the interpretations of synthetized EM responses by the 2-D resistivity model with topographic changes [dashed lines in (c)] and the other models are shown in (d)–(f). (d) 1-D Gauss–Newton inversion. (e) 1-D DNN trained without topographic information. (f) 1-D DNN trained with topographic information.

100 m horizontally and vertically, respectively. EM responses were calculated at 291 points located from 10 to 590 m in the horizontal direction (at 2 m intervals) at a sensor height of 40 m using the 2.5-D forward-modeling algorithm [9]. Compared with the EM responses of the flat resistivity model, topographic changes dramatically affected the EM responses [Fig. 11(c)], and these distortions can be viewed as 2-D variations in subsurface resistivity during inversion. Fig. 10(d) shows a resistivity image inverted using the 1-D Gauss–Newton three-layered model with topographic changes

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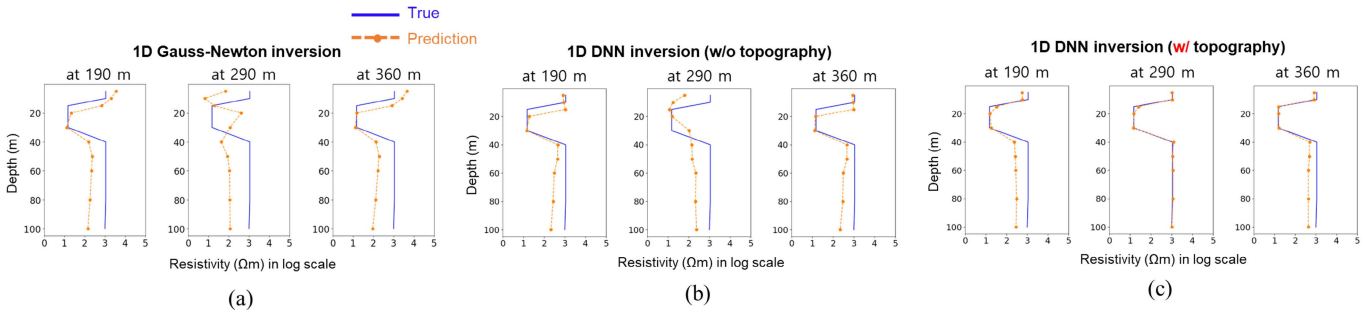


Fig. 12. Comparison of the true (blue lines) and recovered (orange dashed lines with dots) 1-D resistivities at the 190-, 290-, and 360-m distances of the synthetic 2-D resistivity model of Fig. 10. (a) 1-D Gauss–Newton inversion. (b) 1-D DNN trained without topographic information. (c) 1-D DNN trained with topographic information.

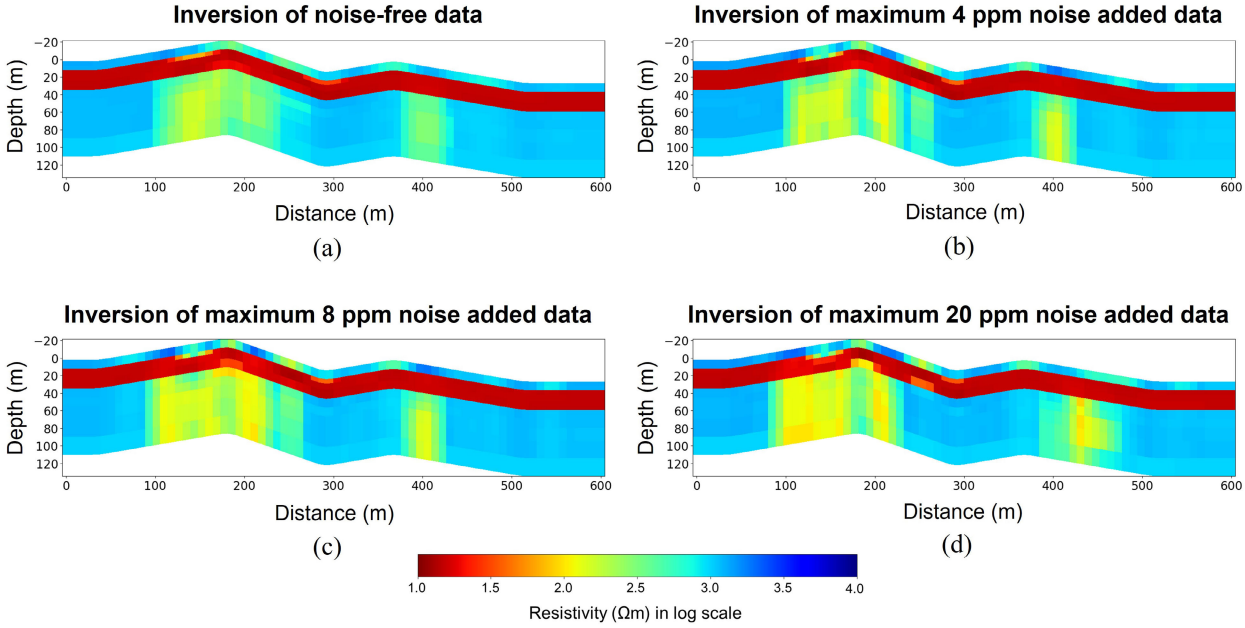


Fig. 13. Interpretation results according to the various intensities of noise levels (EM responses were calculated from the 2-D resistivity model in Fig. 11(b), and the sensor height was fixed to 40 m). (a) Inversion of noise-free data. (b) Inversion of maximum 4-ppm noise-added data. (c) Inversion of maximum 8-ppm noise-added data. (d) Inversion of maximum 20-ppm noise-added data.

[Fig. 11(b)]. The conductive layer below the thin resistive layer is relatively well delineated, but the resistivity of the first layer and the thickness of the second (conductive) layer are not well recovered given the nonuniqueness of 1-D inversion. In addition, since the third layer does not have sensitivity due to conductive upper layer, the inverted resistivity is highly inaccurate and varies by the initial resistivity. Apart from these 1-D inversion effects, the inverted resistivity image shows clear distortions around 190, 290, and 360 m, because the EM responses are contaminated by topographic variations.

To show the importance of topographic changes during DNN inversion, we initially trained the 1-D DNN inversion model without topographic information. This model was trained only using flat resistivity models and their EM responses pairs with the same architecture and hyperparameters as previously described 1-D DNN inversion with topography information. As for 1-D Gauss–Newton inversion, 1-D DNN inversion without topographic information predicted clearly distorted images [Fig. 11(e)] even though the second conductive layer was better defined than after Gauss–Newton inversion. This shows that performance of DNN highly depends on the training dataset. Therefore, if that dataset does not reflect the characteristics of the field, it is difficult to obtain reasonable outputs.

Fig. 11(f) shows the predicted resistivity images of 1-D DNN inversion with topographic information (our new method). Even though inclusion of neighboring topographic changes during training is very simple, artifacts and distortions caused by topographic changes are greatly alleviated. To compare the performances in detail, we plotted the 1-D resistivity structures of true and recovered models at 190, 290, and 360 m, respectively, where the topographies clearly change. As shown in Fig. 12(c), the DNN-based inversion method trained with topographic information showed performance to recover the true resistivities and structures comparing with 1-D Gauss–Newton inversion [Fig. 12(a)] and a DNN trained without topography [Fig. 12(b)].

As we described in Section IV, our DNN model was trained not only with noise-free data but also with random noise-added data to enhance the robustness to the noise. The maximum level of the noise-added data was 2 ppm, because the typical noise level of the DIGHEM-V system is known

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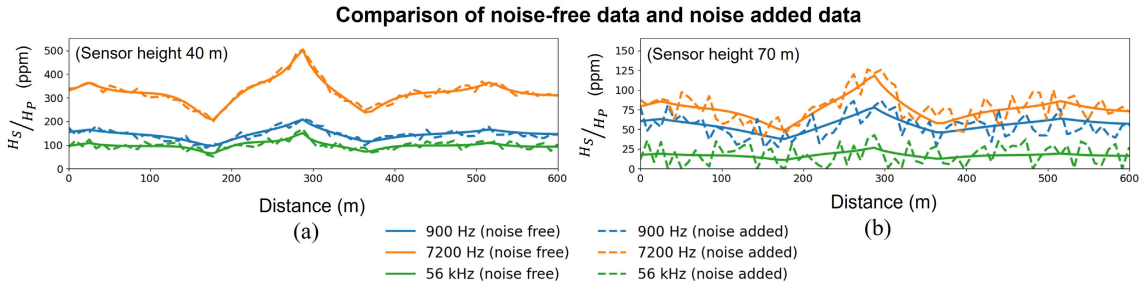


Fig. 14. Comparison of noise-free data and noise-added (maximum 20 ppm) data at (a) 40-m sensor height and (b) 70-m sensor height (bold lines indicate noise-free data, and dashed lines indicate noise-added data).

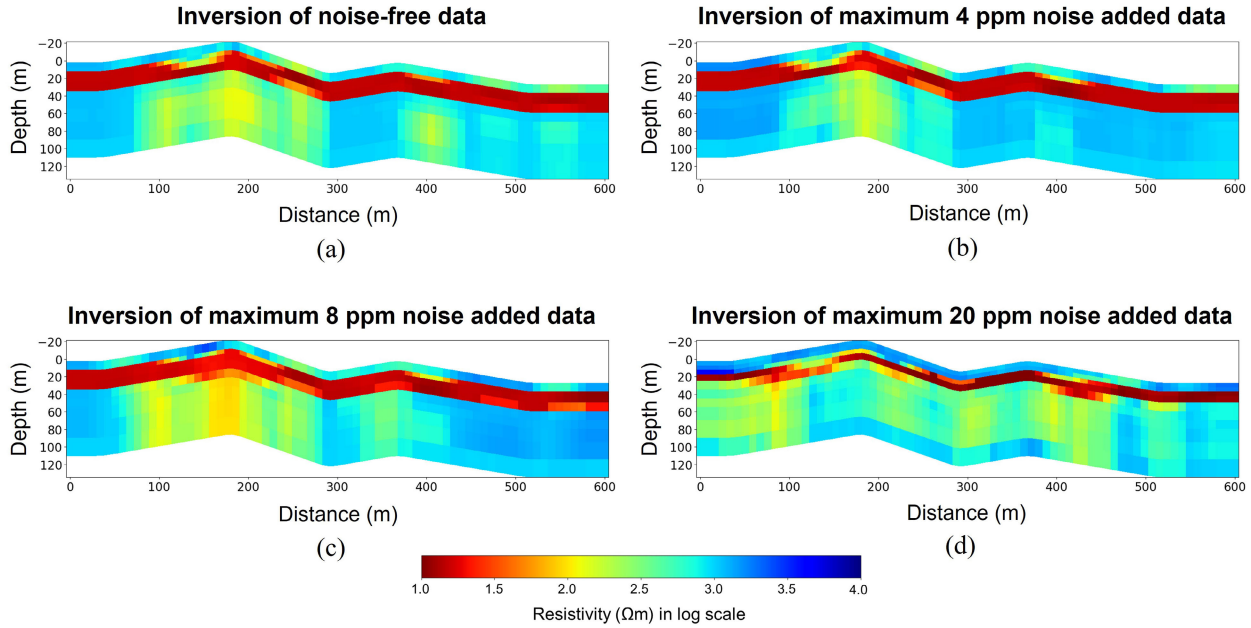


Fig. 15. Interpretation results according to the various intensities of noise levels (EM responses were calculated from the 2-D resistivity model in Fig. 11(b), and the sensor height was fixed to 70 m). (a) Inversion of noise-free data. (b) Inversion of maximum 4-ppm noise-added data. (c) Inversion of maximum 8-ppm noise-added data. (d) Inversion of maximum 20-ppm noise-added data.

to be below 2 ppm. However, in the field survey, there are various situations, such as windy weather or rapid drift of the helicopter, which may lead to the high noise level. Therefore, we tested the robustness of the trained DNN model under the various noise levels for practical use.

We generated uniform random noise with the maximum noise levels of 4, 8, and 20 ppm and added these noises to synthetic data calculated for the resistivity model shown in Fig. 11(b). To test the performance of the trained DNN model on the synthetic data with different random noise levels, we applied our DNN model to four different datasets and presented the prediction results in Fig. 13. As shown in Fig. 13, random noises of different intensities slightly impair the resistivity values and continuity of the predicted model. However, the DNN model can discriminate a clear image of the second conductive layer even when severe noise (20 ppm) was added [Fig. 13(d)]. Unlike conventional 1-D interpretation, which uses a single data point, our DNN model uses three consecutive data points. Therefore, even if each data point is contaminated by substantial noise, our DNN model can reduce the effect of noise by simultaneously using neighboring data points for interpretation. However, at high altitudes above 70 m, the magnitude of the EM responses is relatively reduced due to the low sensitivity, making it susceptible to noise. To see the effect of random noise at high altitude data, we generated EM responses for the same resistivity model with 70-m sensor height and added the maximum 4-, 8-, and 20-ppm noises. Fig. 14 shows the 20-ppm random noise effects to the noise-free data for 40 m [Fig. 14(a)] and 70 m [Fig. 14(b)] sensor heights. As we can see in Fig. 14, noise has a greater effect on data measured at higher altitude. Fig. 15 shows the interpretation results of the DNN according to the noise levels. In contrast to the previous case with a sensor height of 40 m, the predicted resistivity models show more artifacts, and the thin conductive layer was poorly predicted. However, since the predicted results are still reliable below 8-ppm random noise case, we can say that the proposed method can successfully mitigate the random noise effects and enables reliable interpretation for the data measured by the DIGHEM-V system.

VI. Field Application: The Placer Mining Area in Southcentral Alaska

The numerical experiments showed that DNN-based inversion of data from regions with rough topographies improved

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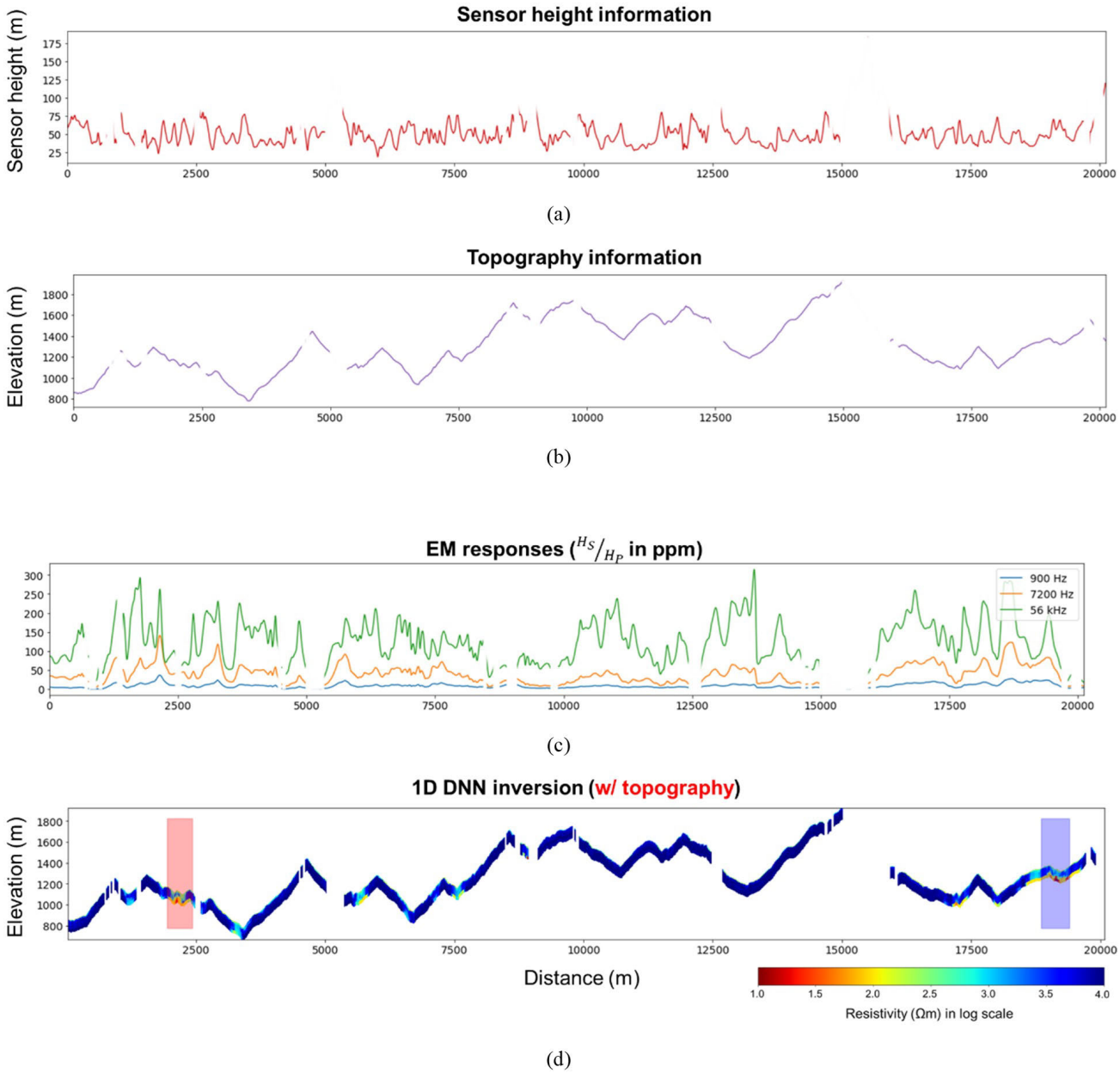


Fig. 16. Input data and results predicted by our 1-D DNN inversion method for the red survey line in Fig. 3(f). (a)–(c) Sensor heights, topographies, and EM responses along the survey line, respectively. (d) Subsurface resistivity images predicted by 1-D DNN inversion. Two regions with conductive anomalies, which will be used to verify the predictions are marked in red and blue.

greatly when topographic information was included in the training dataset. Thus, we applied the DNN-based inversion, trained with topographic information, to a field dataset acquired in Southcentral Alaska (Fig. 3).

Of the many survey lines, Fig. 16(a)–(c) shows the sensor heights, topographies, and EM responses along the red line of Fig. 3(f). The line length is 20 km, and there are over 10 000 data points. However, as mentioned in Section III, data obtained at a sensor height of over 70 m or with steep dips (over ±40°) cannot be handled via DNN-based inversion; such data were deleted during preprocessing (Fig. 16). The trained DNN-based inversion model predicts the 1-D resistivity at the middle point of three data points using the sensor height, topographic information, and EM responses of all three points. Fig. 16(d) shows the concatenated 1-D inversion results for all data points along the line. For the detailed evaluation of inverted results, two conductive regions marked with red and blue rectangles in Fig. 16(d) are investigated and compared with the result of 1-D Gauss–Newton inversion.

Fig. 17(a) shows an enlarged image of the DNN-based inversion results marked in red in Fig. 16(d), and Fig. 17(c) shows the inverted resistivity images of the same data with consideration of only sensor height by 1-D Gauss–Newton inversion. The EM responses of the interpreted resistivity model of Fig. 17(a) (incorporating the topography and sensor height) were calculated using the 2.5-D forward-modeling algorithm [9], and the calculated responses are compared with the field responses in Fig. 17(b). All recalculated EM responses (dashed line) match the field responses (bold line) at all three frequencies, although only adjacent topographic

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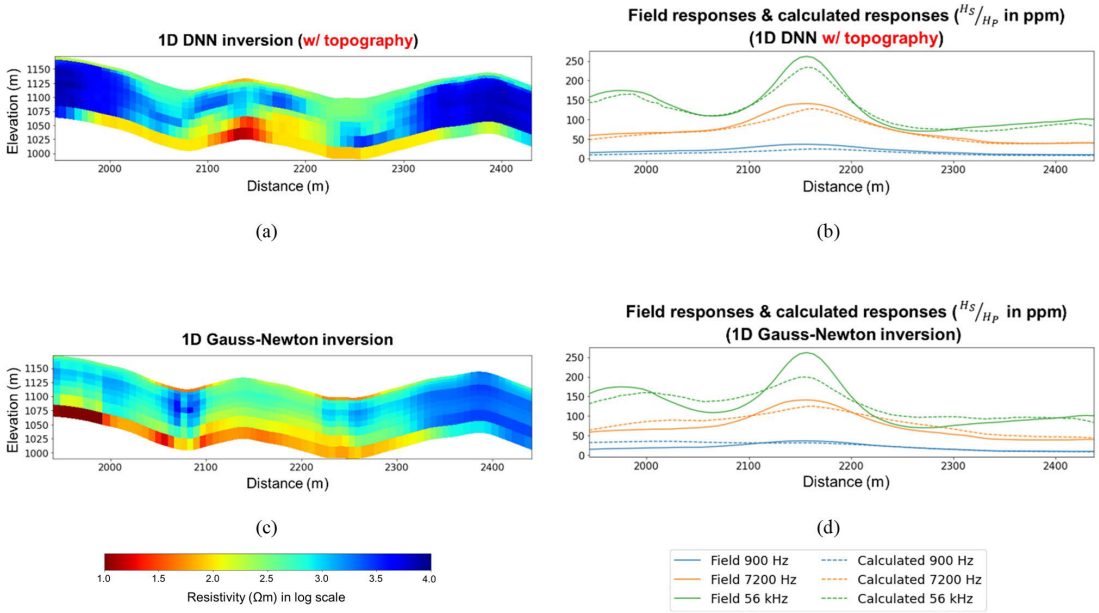


Fig. 17. (a) Enlarged image of the 1-D DNN inversion result marked red in Fig. 16(d). (c) 1-D Gauss–Newton inversion result of the same dataset. (b) and (d) Compare the field data (bold lines) with the synthetic data (dashed lines) calculated using the resistivity models of (a) and (c), respectively, using 2.5-D modeling code that included topographic information.

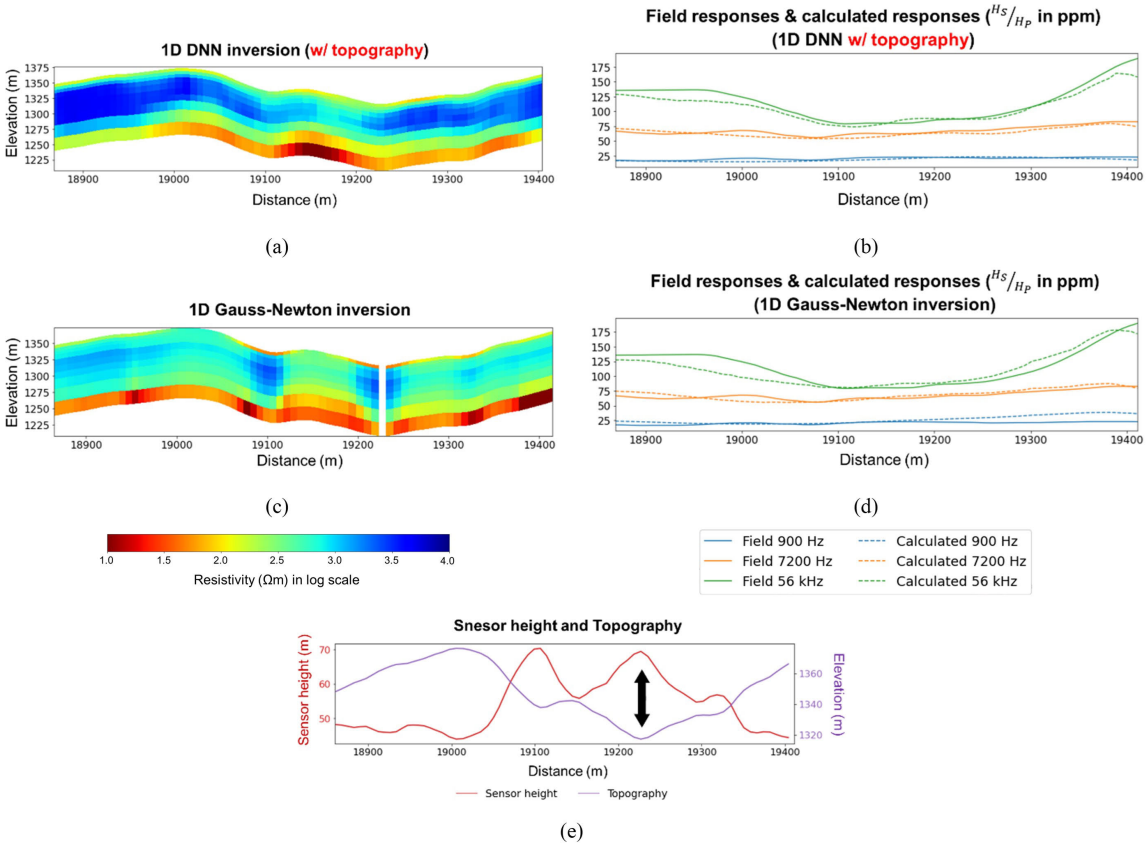


Fig. 18. (a) Enlarged image of the 1-D DNN inversion marked blue in Fig. 16(d). (c) 1-D Gauss–Newton inversion of the same dataset. (b) and (d) Compare the field data with data calculated using resistivity models (a) and (c), respectively (these employ the 2.5-D modeling code and include topographic information). (e) Sensor heights and topographic data for the relevant interval. The black arrow in (e) indicates the zone of inversion failure [blank in (c)].

changes were considered during 1-D inversion. The 2.5-D EM responses for the inverted model of the 1-D Gauss–Newton inversion were next calculated with incorporation of both topography and sensor height.

As shown in Fig. 17(d), unlike our 1-D DNN inversion incorporating topography, the recalculated EM responses of the 1-D Gauss–Newton inversion result differed markedly from the field responses. In particular, the calculated EM

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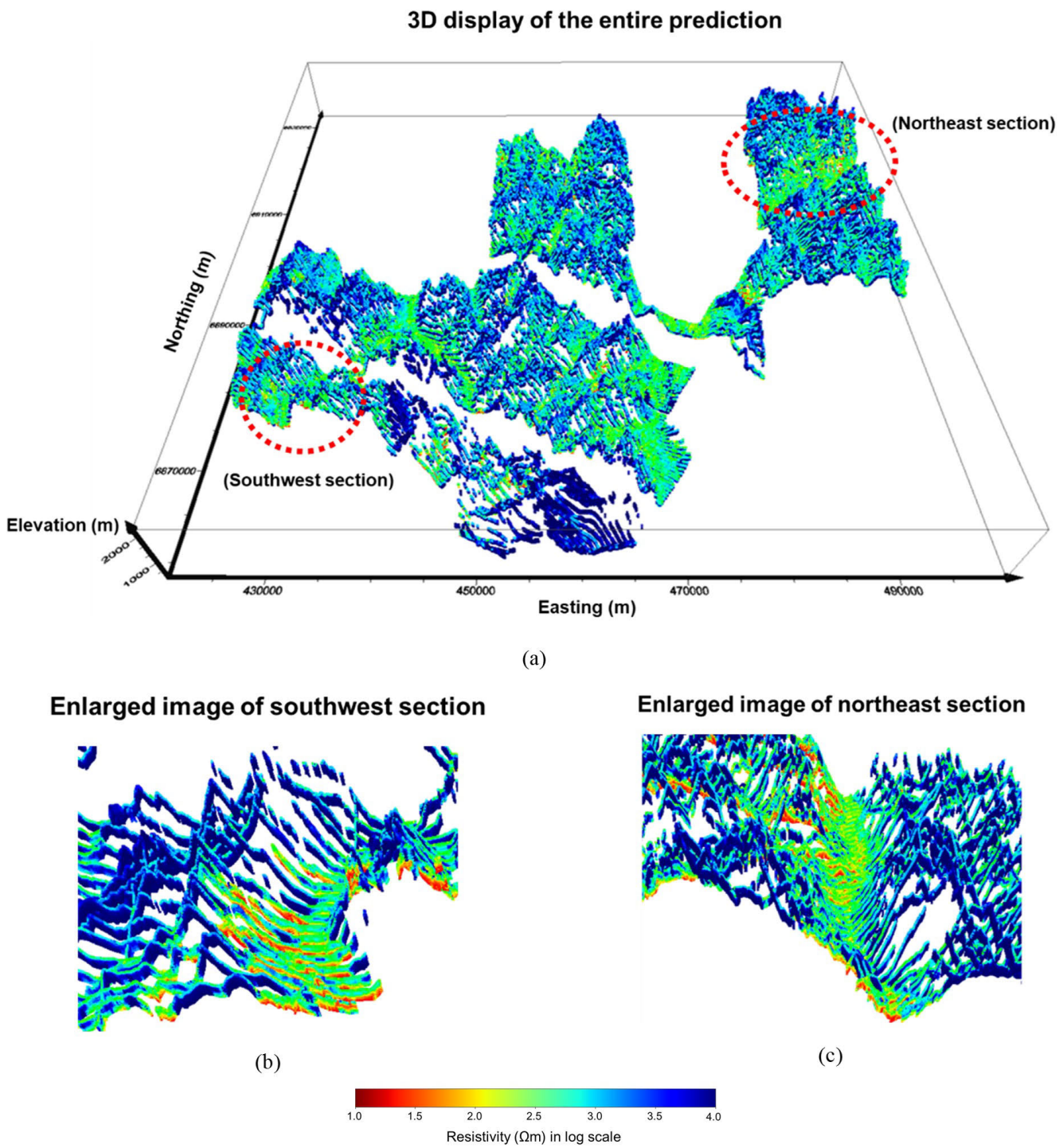


Fig. 19. (a) 3-D display of the predicted results for the entire field dataset. Red circles indicate conductive areas in the southwest and northeast. Enlarged images of (b) southwest and (c) northeast sections.

responses at the highest frequency (56 kHz) (which are more affected by topographic changes) showed large gaps, because 1-D Gauss–Newton inversion minimizes the residual differences in EM responses between resistivity models with and without topographic variation.

For the area marked blue in Fig. 16(d), the interpretation of our trained DNN-based inversion model is shown in Fig. 18(a), and a comparison between the field responses and the recalculated EM responses using the 2.5-D forward-modeling algorithm is shown in Fig. 18(b). The recalculated EM responses matched well with field responses at all frequencies except for small discrepancies due to 1-D interpretation of subsurface resistivity. However, 1-D Gauss–Newton inversion failed in the blank zones of Fig. 18(c), because there are no corresponding 1-D EM responses, which can represent the distortion of EM response due to the topographic change at the corresponding sensor height marked with black arrows in Fig. 18(d).

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Thus, through the analysis of field data application, we could confirm that topographic information must be considered when interpreting AEM data acquired over mountainous terrain. Our simple idea, incorporation of adjacent topographic variations into 1-D inversion, is very effective. In terms of computation time, our DNN model required a few seconds to provide all results for the entire survey line of Fig. 16(d), whereas the 1-D Gauss–Newton method required 3 h to provide each of the interpretations of Figs. 17(c) and 18(c). We applied the trained DNN to the entire field dataset of Southcentral Alaska (Fig. 3). Fig. 19(a) shows the concatenated 1-D inversion results for the entire area, and Fig. 19(b) and (c) shows enlarged images of the red dotted circles in Fig. 19(a). We could recognize continuously connected conductive targets at a glance, which may be related to buried gold mines. Again, the figure was created in 20 min without any additional parameter, whereas conventional 1-D Gauss–Newton inversion would take about three years. Therefore, real-time interpretation of AEM data is possible, because our DNN dramatically reduced computation time and minimized the human interventions.

VII. Conclusion

Conventionally, most AEM data are interpreted using 1-D approaches, because 2-D or 3-D interpretation is very time-consuming and computationally costly. Although 1-D inversion is fast and efficient, the conventional Gauss–Newton inversion cannot accommodate big data. Also, conventional inversion requires sophisticated parameter settings depending on the condition of field data to obtain reliable results. As an alternative, DNN-based inversion techniques enable rapid interpretation of large AEM datasets without any need for additional parameters. However, the existing DNNs do not consider topographical changes that distort EM responses. Since lots of airborne surveys are performed in mountainous areas, it is necessary to consider the topography information in the training process.

We proposed the 1-D DNN inversion, which can alleviate the topography effects that cause distortions of EM responses. As topographic changes are included during training, the model learns what occurs when topographic changes affect subsurface layered-earth resistivities and survey heights. To simulate topographic changes, we used pseudo-1-D modeling to generate the EM responses of 1-D-layered resistivity models inclined on both sides and centered to the midpoint. The three consecutive EM responses, the sensor heights, and the inclination angles (both sides) served as 1-D DNN inversion inputs for prediction of layer resistivities at the midpoints. In addition, in training stage, noise-added dataset was used together with noise-free dataset to reduce the effect of noise in field data.

The performance of the proposed 1-D DNN inversion on the mitigation of topographic effect was presented through numerical experiments. Although EM distortions attributable to topographic changes were modeled using only adjacent points, artifacts in the predicted resistivities were alleviated, and the conductive target layers were well delineated.

In addition, we proved the applicability of the trained DNN inversion to field data through the field dataset applications, which acquired in Alaska for mineral explorations. As this survey covered a very large and very rough area, even conventional 1-D inversion has not been performed on this dataset. Conventional Gauss–Newton 1-D inversion would require approximately three years to interpret the findings even without tuning the inversion parameters. Furthermore, topographic distortions and inappropriate parameter settings may compromise the results. In contrast, the proposed DNN inversion provided the interpretation results for whole dataset in 20 min without any other parameter adjustment. In addition, the prediction results were verified by comparing the recalculated data from the predicted resistivity model incorporating topography with the measured field data. The developed DNN inversion model is designed for the DIGHEM-V system not for a specific field data; it can directly be applied to another dataset acquired using the same instrument. With the introduction of DNN technology, the computation time and human efforts are dramatically reduced, and, by incorporating topographic change in training stage, the artifacts in inverted images due to the topography effect can be effectively reduced.

However, since the developed method is based on a 1-D approach, it has limitation to interpret 2-D or 3-D anomalous bodies, such as fault boundary, or isolate body accurately. Thus, the following high-dimensional interpretation techniques need to be applied for prospecting area detected by our method. However, the proposed 1-D DNN inversion technique enables rapid interpretation and will bring a breakthrough in the practical application of AEM survey. Through this study, we expect that the AEM survey, which has been conducted so far but has not been widely used due to the lack of a practical interpretation tool, will receive more attention.

Appendix: Procedure of 1-D Resistivity Model Generation

Before generating 1-D resistivity models, we first set the maximum number of 1-D models to ensure that we will generate and store the numerical modeling results without overloading our computer memory. We set the maximum number of 1-D resistivity models to 6300, and the total number of models increases to 510 300 when 81 topography patterns are considered.

Next, we set the background resistivity values, because the numerical modeling algorithm calculates the secondary field using Maxwell's equation, which is shown as (A1) and (A2), using the difference of anomalous resistivity and background resistivity

$$\nabla \times \boldsymbol{E}^s + \hat{z}\boldsymbol{H}^s = \boldsymbol{0} \tag{A1}$$

$$\nabla \times \boldsymbol{H}^s - \hat{y}\boldsymbol{E}^s = \Delta\boldsymbol{\sigma}\,\boldsymbol{E}^p \tag{A2}$$

where

$\boldsymbol{E}^p$: primary electric field, V/m;
$\boldsymbol{E}^s$: secondary electric field, V/m;
$\boldsymbol{H}^s$: secondary electric field, A/m;
$\Delta\boldsymbol{\sigma} = \boldsymbol{\sigma} - \boldsymbol{\sigma}_b$;
$\boldsymbol{\sigma}$: anomalous electric conductivity, S/m;
$\boldsymbol{\sigma}_b$: background electric conductivity, S/m;
$\hat{y} = \boldsymbol{\sigma} + i\boldsymbol{\omega\varepsilon}$: impedivity;
$\hat{z} = i\boldsymbol{\omega\mu}$: admittivity;
$\boldsymbol{\varepsilon}$: permitivity, F/m;
$\boldsymbol{\mu}$: magnetic permability, H/m;
$\boldsymbol{\omega}$: angular frequency, Hz.

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Algorithm 1 1-D Resistivity Model Generation Procedure

1) Set maximum number of resistivity models, n_max = 6300
Set backgr ound resistivities, res_bak = [10, 100, 1000, 10000],

2) Set layer boundaries, boundary = [5, 10, 15, 20, 30, 40, 50, 60, 80]

3) Generate empty array to save ID resistivity models.

4) Generate **homogeneous resistivity models**
Iteration loop 1:
  select backgr ound resistivity and save resistivity models.
Count iter ations, itr = itr + 1

5) While (itr <= n_max)
 5-1) Generate **two-layered resistivity models**
  Iteration loop 1:
   select layer boundary
  Iteration loop 2:
   select a background resistivity
   allocate it to the deepest layer
  resistivity of shallow layer = random.uniform[1,10000]
  If resistivities of all layers are different,
   then save resistivity model and count iterations, itr = itr + 1.
 5-2) Generate **three- or four-layered resistivity models**
  Iteration loop 1:
   select layer boundaries
  Iteration loop 2:
   select a background resistivity and allocate it to the deepest layer
  resistivity of remaining layers = random. uniform[1,10000]
  If resistivities of all layers are different,
   then save resistivity model and count iterations, itr = itr + 1.
 5-3) Count duplicates. If there is a duplicate, subtract the iteration number.
  Remove duplicate

We set the resistivity of the deepest layer as a background resistivity and those of the remaining layers as the anomalous resistivities. We used the background resistivity values of 10, 100, 1000, and 10 000 Ωm, which were used in the research of Bang et al. [8], and the anomalous resistivity was selected in the range of 1–10 000 Ωm using a uniform random function.

As mentioned in Section III, we generated from a homogeneous model to a maximum of four-layer models. In the case of homogeneous models, four different models were generated with the background resistivities of 10, 100, 1000, and 10 000 Ωm.

However, when it comes to multilayer models, it is necessary to assign the depth of the boundary between the individual layers. To ensure an adequate sensitivity to depth, we chose the boundary of each layer as one of the predefined values, namely [5, 10, 15, 20, 30, 40, 50, 60, and 80] m. Beyond a depth of 100 m, we used the resistivity value of the background medium.

Here is the pseudocode of the 1-D resistivity model generation to facilitate the understanding of our method.

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